RAGUNATH

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PROFESSIONAL SUMMARY

Electronics & Communication Engineering graduate (2025) with hands-on experience building **embedded vision and IoT systems** — including a **Raspberry Pi-based humanoid robot** with **real-time facial recognition** and servo-driven physical response. Complements this with **6 months of software development internship experience** delivering full-stack and AI-integrated applications. Seeking to bring hardware-software integration skills to embedded and control-systems work.

TECHNICAL SKILLS

- **Embedded & Hardware:** Raspberry Pi 4B, OAK-D Lite depth/vision camera, servo motor control, GPIO & sensor interfacing, edge-device deployment
- **IoT & Connectivity:** Bolt IoT integration, Telegram API automation, real-time device-to-cloud alerts
- **Computer Vision / AI:** Facial recognition, emotion detection, OCR pipelines, model-driven (non-rule-based) inference design
- **Programming:** Python, PHP (Laravel)
- **Backend & Data:** Flask, MySQL, SQLite, FAISS (vector search), Ollama (local LLM inference)
- **Tools & Practices:** Git, Streamlit, Tkinter GUI development, modular application architecture

EMBEDDED & IOT PROJECTS

Humanoid Robot – Embedded Vision & Control System

- Built a humanoid robot platform on a **Raspberry Pi 4B**, integrating an **OAK-D Lite depth camera** with servo motors for physical response
- Implemented real-time **facial recognition and emotion detection** using genuine model-driven inference rather than placeholder logic
- Designed the vision-to-actuation pipeline connecting camera input, on-device inference, and servo motor output

Face Recognition IoT System

- Developed an **OAK-D Lite**–based facial recognition system with a **Tkinter** desktop GUI for real-time monitoring
- Integrated **Bolt IoT** and **Telegram APIs** to trigger remote alerts and automate responses to recognition events

EXPERIENCE

Software Development Intern | ***Dsignz Media, Coimbatore*** 6 Months

- Contributed to a **Laravel**-based internal Workforce Management admin dashboard used for employee and project tracking
- Worked across the **Python and PHP/Laravel** stack on full-stack feature development
- Built and deployed AI-integrated tools (**RAG-based systems**, OCR pipelines) as part of applied AI/ML exploration

EDUCATION

B.E., Electronics and Communication Engineering | ***Muthyammal Engineering College*** Graduated 2025